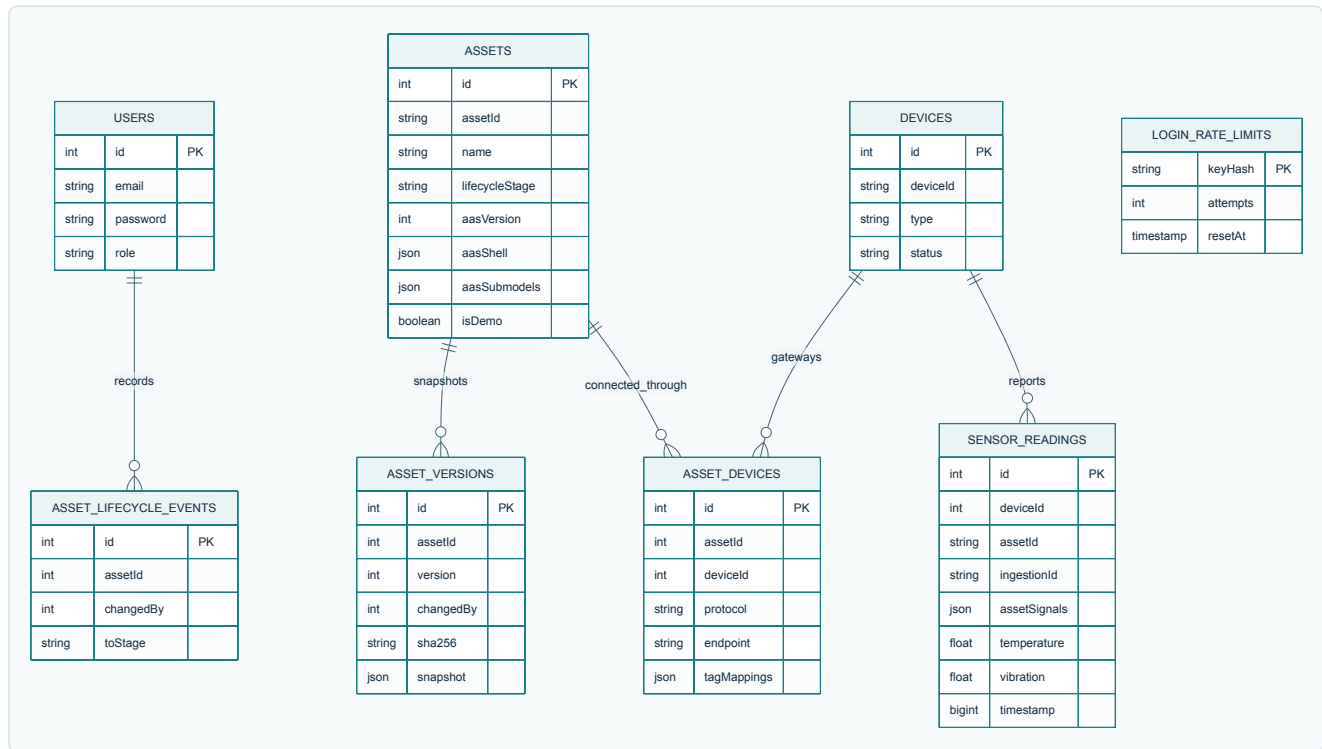
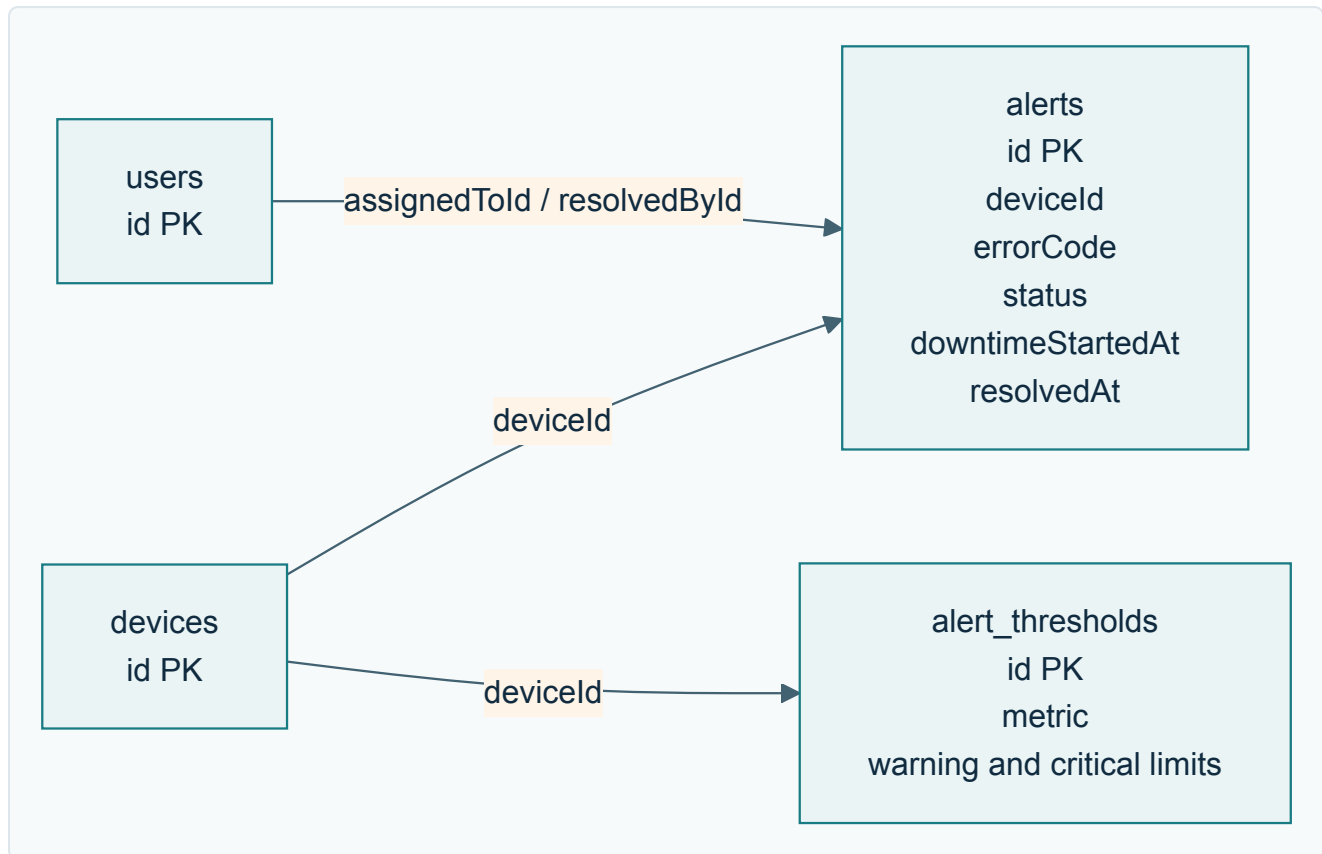


Application Database Schema

The dashboard uses PostgreSQL with Drizzle ORM. The source of truth is `drizzle/schema.ts`; checked-in SQL migrations live in `drizzle/`.



Incident response and downtime records



Threshold-generated incidents use stable SF-* platform classifications. A vendor diagnostic code is stored as supplied when available. The engineer role provides technician assignment and resolution permissions; downtime is explicitly confirmed by the assigned engineer or administrator. A lost device connection does not automatically prove a factory outage. Existing records inferred by migration 0010 need operational review before use in downtime KPIs. Mean downtime-to-resolution uses only records with both downtime and resolution timestamps.

Table	Purpose
users	Local identity, bcrypt password hash in password, role, and sign-in timestamps. Never expose the hash in API responses.
devices	Connectivity devices. New registrations use enum type gateway or edge_device; legacy sensor/actuator/controller values remain readable. Includes status, location, firmware, and simulator marker. Migration 0015_melted_dragon_lord adds edge_device.
assets	Equipment identity, manufacturer name/designation, separate required postal fields, article and order codes, optional serial/rated values, lifecycle, current AAS JSON, monotonic AAS revision, and simulator marker.
asset_devices	Asset-to-gateway protocol, endpoint, and tag/register mappings. Store credentials on the edge gateway, not in mappings.
asset_lifecycle_events	Actor, transition, timestamp, and engineering note audit history.
asset_versions	Append-only AAS snapshots keyed by asset and revision, with change note, actor, and SHA-256 integrity digest. Migration baseline rows have no digest because prior revisions were not recorded.
login_rate_limits	Hashed IP/email identifiers, fixed-window counters, and expiration timestamps shared across application replicas.
sensor_readings	Nullable measurements, AAS asset attribution, named signals, Unix epoch millisecond timestamps, and a unique ingestion digest for retry deduplication. Indexes cover asset/date and device/date queries.
alerts	Stable platform/vendor error code, severity/status, metric observation, acknowledging actor/time, assigned engineer/time, confirmed downtime start, resolving actor/time, and lifecycle timestamps. Per-source transaction locks deduplicate threshold incidents by device/asset/metric; query indexes cover status/date and assignee/status.
alert_thresholds	Enabled per-device warning and critical bands used to classify telemetry into incidents.
firmware_versions, ota_deployments	Firmware catalog metadata and legacy rollout rows. OTA is disabled; a rollout row is not proof that a device downloaded, verified, installed, or booted a release.

Connection and lifecycle ids are application-level relationships, not database-enforced foreign keys in the current schema. Oracle-hosted BaSyx owns separate storage in the Aiven basyx PostgreSQL database; it is not part of the Drizzle application schema. Demo rows are marked isDemo and are API-seeded only when demo data is enabled and no real assets exist.

Review migration SQL before applying changes. Set DATABASE_URL to the intended database before running pnpm db:migrate. pnpm e2e uses its own disposable PostgreSQL container.

Durable incident notifications

Migration 0013_incident_notifications adds notification_inbox and an alert trigger. In the alert transaction, new warning/critical incidents and critical escalations notify current engineers/admins; assignment changes notify new/previous assignees and admins; resolution notifies the assigned

technician/admins. The trigger does not backfill historical incidents or emit duplicate messages on unrelated incident updates.

Each row belongs to one account, links to its alert and retains title/body, read time, email status, attempts, next attempt time, safe error summary and Resend acceptance time. Foreign-key cascades remove its rows when the associated account/alert is deleted. The delivery worker atomically claims due rows using a two-minute lease and retries after a crash. `accepted` means the Resend request was accepted, not that the email reached a mailbox. Eight failed attempts become `failed`; an authorized owner may retry after correcting configuration. `unconfigured` and `no_recipient` remain visible.

The inbox is queryable only by its owner. Reads do not acknowledge/resolve the factory incident. The Resend API key, sender and recipient-domain policy are runtime secrets/configuration, not editable browser notification settings. Plan retention/archive policies for both telemetry and inbox history before long-term production operation.

Migration `0014_account_email_identity` adds a case-insensitive unique email index. Null emails remain allowed for legacy identity records. Existing duplicate case-folded email addresses must be reconciled by an administrator before migration. Production self-registration is closed; further accounts are created by an authenticated administrator after private first-admin bootstrap. Passwords are never included in account API responses.